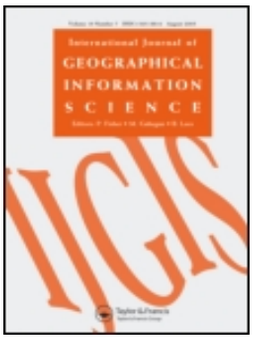

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REVIEW ARTICLE

Influence of positional accuracy, sample size and scale on modelling species distributions: a review

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Species distribution models (SDMs) are an important tool in biogeography and ecology and are widely used for both fundamental and applied research purposes. SDMs require spatially explicit information about species occurrence and environmental covariates to produce a set of rules that identify and scale the environmental space where the species was observed and that can further be used to predict the suitability of a site for the species. More spatially accurate data are increasingly available, and the number of publications on the influence of spatial inaccuracies on the performance of modelling procedures is growing exponentially. Three main sources of uncertainty are associated with the three elements of a predictive function: the dependent variable, the explanatory variables and the algorithm or function used to relate these two variables. In this study, we review how spatial uncertainties influence model accuracy and we propose some methodological issues in the application of SDMs with regard to the modelling of fundamental and realized niches of species. We distinguish two cases suitable for different types of spatial data accuracy. For modelling the realized distribution of a species, particularly for management and conservation purposes, we suggest using only accurate species occurrence data and large sample sizes. Appropriate data filtering and examination of the spatial autocorrelation in predictors should be a routine procedure to minimize the possible influence of positional uncertainty in species occurrence data. However, if the data are sparse, models of the potential distribution of species can be created using a relatively small sample size, and this can provide a generalized indication of the main regional drivers of the distribution patterns. By this means, field surveys can be targeted to discover unknown populations and species in poorly surveyed regions in order to improve the robustness of the data for later modelling of the realized distributions. Based on this review, we conclude that (1) with data that are currently available, studies performed at a resolution of 1–100 km^2 are useful for hypothesizing about the environmental conditions that limit the distribution of a species and (2) incorporating coarse resolution species occurrence data in a model, despite an increase in sample size, lowers model performance.

Keywords: spatial error; grain size; resolution; SDM; uncertainty; potential distribution; realized distribution

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Introduction

It is extremely important in biogeography and ecology to understand the relationship between species and their habitats, because this relationship is fundamentally tied to questions of distribution and abundance (Huston 2002). Species distribution models (SDMs) are an important tool for understanding these relationships and are used for the purposes of both fundamental and applied research. For example, SDMs have been used to evaluate the relative role of environmental factors affecting the distribution of a species (e.g. Storch *et al*. 2003, Gottschalk *et al*. 2007), to assess the potential distribution of invasive species (e.g. Hallstan *et al.* 2009, Chytrý *et al*. 2012), to evaluate the conservation status of endangered species (e.g. Jackson and Robertson 2011), to forecast shifts in the geographic distributions of species under climate change (e.g. Hongoh *et al*. 2012, Renwick *et al*. 2012), to obtain range estimates for assemblages of species (Amboni and Laffan, in press), and to assess the possible impact of changes in land use on species distribution (e.g. Schleupner and Link 2008). Numerous methods can be used to model species distributions (Guisan and Zimmermann 2000, Franklin 2009), often in conjunction with geographic information system (GIS), and many studies have attempted to evaluate their predictive performance (Pearce and Ferrier 2000, Elith *et al*. 2006, Tsoar *et al*. 2007).

Species distribution modelling requires species occurrence data and environmental variables in order to produce a set of rules that identify and scale the environmental space where the species was collected or observed and can further be used to predict the suitability of a site for the species (Araújo and Guisan 2006). Unfortunately, there are many inherent sources of uncertainty in SDMs, which can affect the model performance. Uncertainty arises from differences between the complex, continuous nature of the real world and the simplified, discrete representation of samples in GIS.

Three main sources of uncertainty are associated with the three elements of a predictive function: the dependent variable, the explanatory variables and the algorithm or function used to relate these two variables. Concern about uncertainty in spatial data, its propagation through the modelling processes and its influence on model performance is not new (Heuvelink 1998) and has been widely studied in recent years. Some of the possible sources of uncertainty related to species occurrence data are sample size (Stockwell and Peterson 2002, Hernandez *et al*. 2006, Guisan *et al*. 2007b, Kaliontzopoulou *et al*. 2008, Wisz *et al*. 2008, Reside *et al*. 2011), positional accuracy (Graham *et al*. 2008, Johnson and Gillingham 2008, Fernandez *et al*. 2009, Osborne and Leitão 2009, Naimi *et al*. 2011) and availability of true-absence data (Lobo *et al.* 2010, Lobo and Tognelli 2011). There are also many concerns about uncertainty in environmental variables, for example, selection of relevant environmental variables (Austin 2007, Austin and Van Niel 2011, Synes and Osborne 2011, Williams *et al*. 2012), data origin (Müller *et al.* 2009) and spatial resolution (Ferrier and Watson 1997, Tobalske 2002, Graf *et al*. 2005, Guisan *et al*. 2007a, Heikkinen *et al*. 2007, Kaliontzopoulou *et al*. 2008, Seo *et al*. 2009, Gottschalk *et al*. 2011). Despite the well-known limitations of the spatial data used in SDMs and their impact on model interpretation, many analysts use spatial data non-critically, without regard to these issues (Hunter *et al*. 2009). Moreover, many studies appear to use only data that are readily at hand, without a detailed examination of its quality or an explanation of the relevance of the selected environmental variables (Elith and Leathwick 2009, but see Williams *et al*. 2012). Further research on this topic is therefore warranted (Graham *et al*. 2008, Austin and Van Niel 2011, Hanberry *et al.* 2012).

Due to the extensive use of SDMs and increasing awareness and understanding of uncertainty in spatial data, the attention given to this issue has increased gradually over the last decade. Several reviews have contributed to this topic. Regan *et al.* (2002) developed a

taxonomy of uncertainty for ecology, and Elith *et al.* (2002) used this taxonomy to outline common sources of error in habitat modelling, along with a review of aspects of uncertainty and suggested methods for investigating and mapping uncertainties. Jager and King (2004) proposed a taxonomy of approaches to clarify the relationship between spatial uncertainty and ecological model predictions, and Barry and Elith (2006) reviewed errors resulting from data deficiencies and errors introduced by model specification. Heikkinen *et al.* (2006) reviewed several factors which may contribute to uncertainty in SDMs, with emphasis on the impact of these uncertainties on the predictions of the potential species distributions under climate change. Recently, Rocchini *et al.* (2011) reviewed the most important sources of uncertainty, which formed during species distribution data acquisition procedures.

Species occurrence data and environmental explanatory variables that have been obtained recently tend to be accurately located (Franklin 2009) and, as briefly outlined above, the number of publications on the performance of several correlative modelling procedures, which aim to estimate the potential or realized distributions of species, is growing exponentially. In this study, we review how spatial uncertainties arising from spatial deficiencies in species occurrence data and environmental explanatory variables influence model accuracy, and we examine the main methodological issues in the application of SDMs with regard to modelling the realized and potential distribution of species. More specifically, we discuss positional accuracy, sample size and scale, which are inextricably linked. Our review here is a contribution to knowledge in relation to the 'data model' for SDMs (Austin 2002, 2007, Williams *et al*. 2012).

Modelling the potential distribution versus the realized distribution

To create an accurate SDM, it is necessary to distinguish clearly between potential distribution and realized distribution models (sensu Jiménez-Valverde *et al.* 2008). The range of environmental conditions that species are physiologically able to tolerate defines their potential distribution in geographical space (ecological niche models sensu Soberón 2010). This has also been termed the physiological niche and relates to the fundamental niche of a species (Austin *et al*. 2009, Soberón and Nakamura 2009). Due to other factors such as competition or disturbance, species typically occupy a subset of their potential distribution. This is termed the realized distribution (distribution models sensu Soberón 2010) and is also known as the actual distribution (Soberón and Nakamura 2009). Importantly, both concepts – potential and realized distributions – refer to a particular moment or a discrete period in time (Soberón 2007), and a mismatch between the time ranges of species occurrence data and environmental variables can result in poorer distribution models (Roubicek *et al*. 2010). The occurrence data and environmental variables should have comparable temporal resolution.

We suggest that this theoretical concept should be particularly followed when the influence of possible uncertainties (e.g. positional error) is tested. For example, in a study by Graham *et al.* (2008), when only climatic and topographic variables were used, the performance of the models was better with degraded species occurrence data with an artificially incorporated random 5 km shift than with the original data. On the other hand, models which additionally included soil, vegetation and disturbance variables performed better with the original data than with the degraded data. At finer resolution, species are often absent from a suitable habitat. For example, species may be absent because of dispersal limitation (Kadmon and Pulliam 1993), biotic interactions (Araújo and Luoto 2007, Chee and Elith in press) and historical events (Syphard and Franklin 2010). Models need

to include variables related to these stochastic or ecological processes, allowing a more comprehensive model of the realized niche to be developed. Moreover, an appropriate validation of the realized distribution requires true-absence data (Václavík and Meentemeyer 2009). The use of pseudo-absence data generates different model predictions in the gradient between potential and realized distributions, depending on the method for selecting the pseudo-absence data (Chefaoui and Lobo 2008). Models that exclude these fine-scale determinants of distribution, and consider only the 'main effects' related to physiological responses, such as suitable climate and soils, provide a partial expression of the potential distribution of a species (Jiménez-Valverde *et al.* 2008). In addition, arbitrary selection of different sets of climatic variables, without a clear ecological rationale, results in variation in the modelled distribution of species (Synes and Osborne 2011).

Species distribution data uncertainty

Positional accuracy

Species distribution modelling assumes that species occurrence data are accurate in their location. Most spatial data are now collected using global positioning system (GPS) technology with fine positional accuracy, and recent species occurrence records can have positional accuracy down to 1 m, if the satellite signal is not attenuated by site conditions. Recent devices for tracking animals with GPS equipment integrated into them (e.g. into a collar) are often biased, because vegetation and terrain may interfere with the satellite signal (Frair *et al*. 2004). Moreover, a vast amount of species occurrence data in museums, atlases and natural history collections often lacks the desired geographic accuracy (Graham *et al*. 2004, Newbold 2010). For example, retrospectively geo-referenced museum data collections have uncertainty associated with lack of geographic details in the textual descriptions (Wieczorek *et al.* 2004). Automated processes have been developed for georeferencing species occurrence data and assigning accuracy attributes to them (Guralnick *et al*. 2006, Hill *et al*. 2009). Knowledge of location sampling uncertainty enables the inclusion of covariate predictors to be tested in the model or to up/down weight the importance of a site and can be devised to help account for bias in SDMs (see Carroll *et al*. 2006). However, to the best of our knowledge, no studies have attempted to use positional error directly as a covariate or a weight in the models.

Positional error can mischaracterize the conditions under which a species occurs, particularly in areas with rapidly changing conditions (e.g. mountainous areas) and can lead to overestimation of the species habitat areas (Feeley and Silman 2010). Obviously, the scale of the study is an important determinant of the effect of location positional error on SDM performance (Osborne and Leitão 2009). At finer resolutions, there is a risk of exacerbating the influence of positional error in the geographical location of a species on SDM performance. For example, species occurrence records can be located in a cell representing a different habitat from the one where the species actually occurred (Guisan *et al*. 2007a). The influence of the magnitude of the positional error on SDM performance must be considered, particularly in relation to the resolution and extent of the study. It is therefore advisable to establish the appropriate plot size for sampling the species occurrences for particular species and purposes (Pandit *et al*. 2010).

Some recent studies have made an effort to use accurate species occurrence data gathered with GPS to quantify the influence of artificially incorporated uncertainty on model performance. Graham *et al.* (2008) carried out their study with a cell size of 1 km^2 and 0.01 km^2 for some regions, and incorporated locality spatial uncertainty (positional error) by using a random 5 km data shift radius. They conclude that certain methods, such as

maximum entropy (Maxent) algorithm (Phillips *et al.* 2006) and boosted regression tree (BRT) techniques are robust to positional error. To establish the relationship between the magnitude of the spatial uncertainty and SDM performance, Osborne and Leitão (2009) randomized the location in various resolution intervals (0–1, 2–3, 4–5 and 0–5 km) and found that model runs with degraded species occurrence data resulted in less accurate models for most cases, but the effect on model performance was small. However, the conclusions from both studies are limited because they used presence-only species data and additionally evaluated the predictive performance of the models using the area under the receiver operating characteristic curve (AUC), which has been seriously questioned (e.g. Golicher *et al*. in press, Jiménez-Valverde 2012). Fernandez *et al.* (2009) measured the performance indirectly, and Johnson and Gillingham (2008) used presence–absence species data. Fernandez *et al.* (2009) simulated extreme positional error by randomizing the location in various intervals (0–5, 5–10, 10–25 and 25–50 km). They showed that with increasing interval (uncertainty) the results are less consistent, though the modelling method has a more than 10-fold larger effect on the results. Johnson and Gillingham (2008) performed their study in finer resolution (cell size 0.062 ha) and incorporated positional error by shifting the animal location in 50 m increments from 50 to 1000 m. They found that positional error was the greatest contributing factor to reduced accuracy of the results. The different results for the influence of positional uncertainty in species occurrence data on prediction accuracy may be explained through increasing spatial autocorrelation in the predictors, which reduces the impact of positional uncertainty on the predictions (Naimi *et al*. 2011).

According to the findings of Graham *et al.* (2008) and Fernandez *et al.* (2009), modelling techniques are not equally influenced by positional error. However, these authors disagree on the sensitivity of certain methods. Similarly, Naimi *et al.* (2011) assessed model performance through AUC and kappa statistics. They concluded that there is a difference in the performance of the modelling methods, and that presence–absence methods performed better than presence-only methods.

It is clear that improvements in data quality related to positional certainty in the location of species occurrence data and the spatial resolution of predictors may lead to major advances. However, there is still intrinsic error in measured phenomena resulting from inherent uncertainty existing in the real world. To avoid the bias resulting from the use of real data, we encourage the ongoing use of artificial species for further study of the influence of positional error on SDMs (e.g. Naimi *et al*. 2011), which offer perfect knowledge of relevant environmental variables and are able to provide an exact measure of the performance of the models (Hirzel *et al.* 2001, Meynard and Quinn 2007, Santika and Hutchinson 2009). We also encourage the development and use of sampling covariates or weights on response locations directly in predictive models to explain some of the error associated with positional accuracy, where this is known.

Sample size

The accuracy of SDM performance also relies on the sample size used to build the models. The sample size used in ecological studies varies from just ten to thousands of cases, and there is concern about the minimum number of records required for effective modelling and how these records are distributed in environmental predictor space (e.g. Pearson *et al*. 2007, Wisz *et al*. 2008). It is important to study how sample size affects model predictive performance, and to identify techniques that are robust to variations in sample size. The sample size should be set to meet the objectives of the study (Foody 2011). In general, the

broader the suite of explanatory variables and the more complex the responses (in terms of shape and interaction), the more data are required to construct a reliable model, and vice versa (Barry and Elith 2006).

The predictive accuracy of methods is clearly related to sample size, and predictions based on a small number of records are unlikely to be as good as those based on a large number of samples (Pearce and Ferrier 2000, Stockwell and Peterson 2002). Each sample size event (presences or absences) is important in representing the environmental gradients, regardless of the relative size of the event (Jimenez-Valverde and Lobo 2006, Jiménez-Valverde *et al.* 2009). However, if sampled by strata, one can deliberately aim to sample the gradients representatively within the ecological range of a species, and this will result in less bias for fewer samples (Austin and Heyligers 1989, Wessels *et al*. 1998). Unfortunately, the amount of data available is limited for many taxa and regions, and the modelling of rare or sparsely sampled species is challenging (Raxworthy *et al*. 2003, Pearson *et al*. 2007, Rebelo and Jones 2010). For example, decreasing the number of species occurrence data required for SDM from 200 to 10 would increase the number of bird species in Mexico that could possibly be modelled more than sixfold (Stockwell and Peterson 2002). In better cases, a choice can be made between many low-precision records and a smaller number of high-precision records. According to Engler *et al.* (2004) and Kaliontzopoulou *et al.* (2008), models based on high-precision data performed better than those based on lower-precision data, even if the sample sizes are higher for the lower-precision data. Similarly, Reside *et al.* (2011) found that incorporating coarse resolution historic species occurrence data to increase data sample size lowered model performance. With regard to uneasy detection of models with positional error in species occurrence data judged by AUC (Osborne and Leitão 2009), it is advisable to use data filtering methods, although this results in fewer records and potentially fewer species that can be modelled (Feeley and Silman 2010).

Although recent studies have shown that prediction success generally increases with sample size, these studies have also revealed the possibility of developing models with small sample sizes for some species (Hernandez *et al*. 2006, Guisan *et al*. 2007b, Wisz *et al*. 2008). Ideally, a predictive system that converges quickly to the greatest accuracy would be preferred. According to Stockwell and Peterson (2002), machine-learning methods reached nearly the same accuracy when there were around 50 samples as with 100 samples. Similarly, Wisz *et al.* (2008) emphasized that no algorithm predicted consistently well with sample size lower than 30 samples. However, both studies used 100 samples at most per species. Hanberry *et al.* (2012) compared models with sample sizes ranging from 30 to 2500, and observed that models with a sample size of 200 and lower tend to be inaccurate. Only the study of Jiménez-Valverde *et al.* (2009) used artificial species and found that, once the model reaches a value of around 70 samples, its reliability measured by AUC became independent of sample size, compared with models with sample size lower than 200 samples. Investigations of possible trade-offs between model performance and sample size indicated that the Maxent algorithm (Phillips *et al.* 2006) is quite robust to variations in sample size and produces useful results with sample sizes as small as 30 occurrences (Hernandez *et al*. 2006, Guisan *et al*. 2007b, Kaliontzopoulou *et al*. 2008, Wisz *et al*. 2008).

A consistent outcome of these studies is that geographical predictions developed from small numbers of occurrence records will be of limited accuracy. Small sample sizes allow only the detection of relatively large differences (Foody 2011). Moreover, data are often spatially and temporally aggregated, and more records are required to accurately map the actual limits of the ranges of a species (Feeley and Silman 2011). Models developed with

small sample sizes should be interpreted as identifying regions with similar environmental conditions to those where the species is known to occur, and not as predicting the actual limits of the range of a species (Pearson *et al*. 2007). Nevertheless, the results are encouraging for identifying potentially suitable areas for species, or for targeting field surveys to discover unknown populations and species in poorly surveyed regions (e.g. Raxworthy *et al*. 2003, Funk and Richardson 2005, Pearson *et al*. 2007, Hernandez *et al*. 2008). This increases the relevance of sparse data in museum collections (Graham *et al*. 2004, Newbold 2010). For example, Ferrier (2002) designed a general framework for improving and iteratively modelling species information by coupling it with the identification of knowledge gaps in survey data toward improving the robustness of the data for later modelling of the realized distributions.

Explanatory variables data uncertainty

Scale

Most ecological processes occur over a limited range of spatial areas (spatial scale) and temporal duration (temporal scale). As a consequence, understanding the dynamics of a particular process (or a phenomenon such as the distribution of a species) requires measurements of explanatory variables on appropriate spatial and temporal scales (Huston 2002). Spatial scale governs what environmental variables can be hypothesized to operate in determining species distribution and abundance (Pearson and Dawson 2003). A mismatch in scale between species occurrence and the explanatory variables may result in a clear decrease in the performance of the model. Scale issues must therefore be handled properly, and possible uncertainties must be described when implementing the models and their predictions in practice (Knegt *et al.* 2010). Spatial scale is a key characteristic of the data, and determines the purpose to which the data can be applied. The intrinsic error in the species observation locations should be lower than the resolution of the environmental variables. However, the choice of spatial scale in the explanatory/environmental data depends on the resolution of the available spatial data that are of relevance to the prediction.

In compiled ecological data, the spatial scale of the sample location may vary, as may the sampling methodology applied in detection, and may have an important effect on model accuracy (e.g. Reese *et al*. 2005, Braunisch and Suchant 2010, Pandit *et al*. 2010). To account for variation in detection or in sampling method between surveys, it is desirable to include covariates which describe contextual or methodological differences in detecting the presence of a species as explanatory variables (e.g. MacKenzie *et al*. 2002, Schmidt 2005). Alternatively weights can be attributed to the response to up- or down-weight the importance of particular locations, whichever is more appropriate (e.g. Carroll *et al.* 2001).

The spatial scale of the environmental data is determined by the resolution (grain size) and the extent of the study area (Morrison and Hall 2002). It is critical to know whether the grain size of the study and the extent are appropriate for the phenomena being studied; otherwise, the results may be misleading (Guisan and Thuiller 2005). It is likely that different species respond to their environments at different spatial scales (Holland *et al*. 2004, Graf *et al*. 2005) and that these scales may be related to the home ranges of the species (Addicott *et al*. 1987). Moreover, the modelled relationships between the dependent variable and the environmental explanatory variables changes over space (Osborne *et al.* 2007, Miller and Hanham 2011).

Environmental variables with coarse resolution lead to a bias against rare environmental entities. For example, the distribution of species may be dependent on such rare environmental entities as small or temporary water bodies (e.g. habitat for birds

and amphibians), which cannot be detected using coarse resolution data but do exist in the natural world. In general, finer spatial resolution environmental data lead to better predictive models, assuming that the data quality associated with the response variable is adequate and does not lead to truncated distributions (Huston 2002). Some authors have compared environmental data sets of land cover variables of different resolution derived from remote sensing platforms and have found no difference in their explanatory power (Seoane *et al.* 2004, Venier *et al.* 2004). This is apparently because the resolution of explanatory variables in the study of Seoane *et al.* (2004) fell within a relatively fine class (30–250 m). The resolution of the response was set to correspond with the resolution of explanatory variables from 857 survey data points. In the study of Venier *et al.* (2004), the resolution of the explanatory variables was fine (200 m and 1 km) relative to the resolution of the response at 5–10 km^2.

Several studies have coarsened the resolution of data to examine its influence on SDM performance. Coarsening the resolution can diversify nearby environmental variables and cause the model to identify spurious combinations of suitable environmental variables for a species, and/or the presence and absence pattern of species may be influenced. For example, Ferrier and Watson (1997) used fine and coarse environmental data to evaluate the effect of spatial resolution (grain size) on SDM performance. The study used 'fine-scale' data for a 4-ha grid and a 'coarse scale' with the same variables generalized to a 25-km^2 grid, and they found degraded model performance using the 'coarse scale'. Guisan *et al.* (2007a) conclude that 10-fold (1–100 km^2) coarsening of resolution does not substantially change the performance of SDMs. However, Kaliontzopoulou *et al.* (2008) reported degraded model performance for 10 km^2 grain size in comparison with 1 km^2 grain size. By contrast, Tobalske (2002) observed better model prediction using their coarse scale (4 km^2) than when using their fine scale (1 km^2). In addition, Graf *et al.* (2005) increased the resolution stepwise from 1 ha up to ~11 km^2 and found that the predictive power was highest using the intermediate scales (~1 and ~2 km^2). However, Heikkinen *et al.* (2007) developed more accurate models for resolutions of 1 and 25 ha than for 1 km^2. The most comprehensive study carried out recently (Seo *et al*. 2009) examined SDM dynamics across a 64-fold (1–4096 km^2) change in resolution and found that the decline in model performance generally accelerated between 64 and 256 km^2 grain size.

From the papers cited here, it seems that the widely used 1–100 km^2 grain size is ideal for identifying the environmental range limits of species and is thus appropriate for modelling species distributions with currently available data. However, the appropriate resolution will differ according to the species under study, the landscape of interest and the environmental variables of relevance to the study. Moreover, these studies did not use grain size finer than 1 ha and, according to the recent findings of Gottschalk *et al.* (2011), remote sensing data with finer spatial resolution are most promising. Additionally, all studies were performed on real data. With regard to their contrasting findings, we suggest that artificial data may be helpful for identifying and resolving the reasons for the differences. It is a matter for further research to investigate whether the lost information from using coarse resolution environmental variables in SDM inevitably degrades the predictive performance of the model.

Conclusions

As this review has shown, positional accuracy, sample size and scale are complex, closely interlinked themes, which unquestionably limit the accuracy of SDMs and should be considered at the outset of any modelling process. With currently available species distribution data, which may contain inaccuracies (e.g. positional error), and environmental variables

with spatial resolution usually in the order of several hundred meters or more, the current literature suggests that studies performed at resolution of 1–100 km^2 over large areas are suitable for identifying the environmental range limits of species and are thus appropriate for modelling species distributions. It is important to note that this recommendation will probably change when more accurate and finer resolution data become readily available. From the papers cited in this review, it is clear that incorporating coarse temporal and spatial resolution species occurrence data in a model lowers performance, despite an increase in sample size.

With regard to modelling the realized and potential distributions of species, we distinguish two cases suitable for different data accuracy. For modelling the realized distribution of a species, particularly for management and conservation efforts which require uncertainty to be minimized, we suggest using spatially accurate species occurrence data with large sample sizes. Accurate species occurrence data can be obtained through appropriate data filtering. An examination of spatial autocorrelation in predictors will help to minimize the possible influence of positional uncertainty in species occurrence data and to avoid wrongly targeted conservation efforts. Additionally, covariates that describe variation in positional accuracy or contextual or methodological differences in detecting the presence of a species can be included as predictors (explanatory variables) or used to generate weights applied to the response variable, whichever is most appropriate. However, if the data is sparse, models to support the generation of hypotheses about the potential distribution of species can be created using a relatively small sample size. Modelling of the potential distribution of species should be directed towards predicting the regions that best meet the minimum physiological requirements of the species. Generally, such data suggest species from poorly surveyed regions, for which models can help to target field surveys to discover unknown populations.

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